DO NOT OPEN THIS BOOKLET UNTIL YOU ARE INSTRUCTED TO DO SO

ECE 3331 - Exam 1 – Spring 2026

I agree to abide by the provisions of the University of Houston Academic Honesty Policy while taking this exam.

Signature:_____

Date:_____

INSTRUCTIONS:

(1) Read and sign the Academic Honesty statement above. Unsigned exams will not be graded.
(2) Including this cover page, this exam has 9 + 1 (blank) = 10 pages. **Do NOT look at the questions**!
(3) This exam is open book and open notes but only written materials are allowed. No electronic aids are allowed.
(4)Communication devices of any kind (eg. cell phone, pager, etc.) may not be used during the exam. Turn them off or give them to the room monitor. No such devices should be visible on your desk.
(5)The use of a four function calculator ONLY is permitted but probably not required.
(6)You will have 80 minutes for this exam.
(7 When time is called you must stop writing immediately and stand up. Continuing to work after time is called will result in substantial penalty (-3 points / min. for 1st 3 minutes, -6 /min. thereafter).

Question	Points	Maximum
1		7
2		8
3		8
4		7
5		10
6		10
7		13
8		13
9		14
10		10
11		20
TOTAL:		100 + 20 Possible Bonus

Question 1 (7 points)

```c
#include <stdio.h>

int main() {
    int i;
    int sum = 0;

    for(i = 1; i <= 6; i++) {
        if(i % 2 == 0)
            continue;
        sum += i;
    }

    printf("%d\n", sum);
    return 0;
}
```

What is the output?

Question 2 (8 points)

```c
#include <stdio.h>

int main() {
    int a = 7, b = 2, c = 3;
    double x = a / b * c;
    double y = (double)a / b * c;

    printf("%.1f\n", x);
    printf("%.1f\n", y);
    return 0;
}
```

What is the output?

Question 3 (8 points)

What is the output?

```c
int main() {
    int a = 2, b = 3, c = 4;
    int x = a++ + --b * c;
    int y = (a > b) && (c++ > 4);

    printf("%d %d %d %d\n", a, b, c, x);
    printf("%d\n", y);
    return 0;
}
```

Question 4 (7 points)

```c
int main() {
    int i = 0;

    while(i < 5) {
        printf("%d ", i);
        if(i == 3)
            break;
        i++;
    }

    return 0;
}
```

What is printed?

Section 2 – Short Programming (33 points)
Write concise programs (5–12 lines).

Question 5 (10 pts)
Read a positive integer n.
Print all multiples of 3 from 1 to n using a for loop.
Example (n = 10):
3 6 9

Question 6 (10 pts)
Read an integer. Print:

- "Positive Even"
- "Positive Odd"
- "Negative"
- "Zero"

Use if statements.

Question 7 (13 pts)
Read a positive integer.
Print:

1. Number of digits
2. Sum of digits

Example:
Input: 523
Output: 3 10

Section 3 – Find and Fix the Errors (28 points)

Rewrite the corrected program.

Question 8 (13 points)

```
#include <stdio.h>

int main() {
    int i;

    for(i = 0; i < 5; i++)
        printf("%d ")

    return 0;
}
```

Fix all errors.

Question 9 (14 pts) — Find & Fix Errors (with precedence trap). You may mark on the question

```
#include <stdio.h>

int main() {
    int x = 10;
    int y = 5;

    /* Intention:
       Print "Match" if (x equals y) OR (x is 10 AND y is 5).
       Otherwise print "No Match".
    */

    if (x = y || x == 10 && y = 5) {
        printf("Match\n")
    } else {
        printf(No Match\n);
    }

    return;
}
```

a) Identify **at least 5 distinct errors** (syntax and/or conceptual).
b) Rewrite the corrected program so it **compiles** and matches the stated intention exactly.

Question 10 (10 points)

```c
#include <stdio.h>

int main() {
    int a = 84, b = 30, t;

    while (b != 0) {
        t = a % b;
        a = b;
        b = t;
    }

    printf("%d\n", a);
    return 0;
}
```

a) What does this program compute (in words)?
b) What is the output?
c) Why does the loop terminate?

Question 11 (20 points)
Problem:
Read a positive integer n and compute the sum of all integers from 1 to n that are NOT divisible by 2 and NOT divisible by 3. For instance, if n = 10, it will sum 1+ 5+7 = 13
A correct but inefficient solution is given:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int sum = 0;
    int i, j;

    for (i = 1; i <= n; i++) {
        int div2 = 0, div3 = 0;

        for (j = 1; j <= i; j++) {
            if (j * 2 == i) div2 = 1;
            if (j * 3 == i) div3 = 1;
        }

        if (!div2 && !div3)
            sum += i;
    }

    printf("%d\n", sum);
    return 0;
}
```

Rewrite the program to make it more efficient in **three distinct ways**:

1. Replace the divisibility logic.
2. Improve control flow.
3. Reduce unnecessary work.